

CoRoT-2b

R-0189

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_180829 · created 2026-07-16T20:53:02+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458359.72766 +/- 0.00099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.217 +/- 0.013
Transit depth (Rp/Rs)^2	4.69 +/- 0.54 [%]
Orbital Inclination (inc)	84.18 +/- 0.85
Airmass coefficient 1 (a1)	1.01 +/- 0.017
Airmass coefficient 2 (a2)	-0.003 +/- 0.019
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	#3 - [243, 253]
Optimal Aperture	3.51
Optimal Annulus	15.5
Transit Duration (day)	0.0836 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.217 $\pm$ 0.013 vs 0.1667 (deviation 2.8 σ)
Duration fit vs published	0.0836 d vs 0.095 d (deviation 1.4 σ)
Mid-time O-C	4.7 min
Depth SNR	12.1
Residual scatter	1.71 %

Light curve

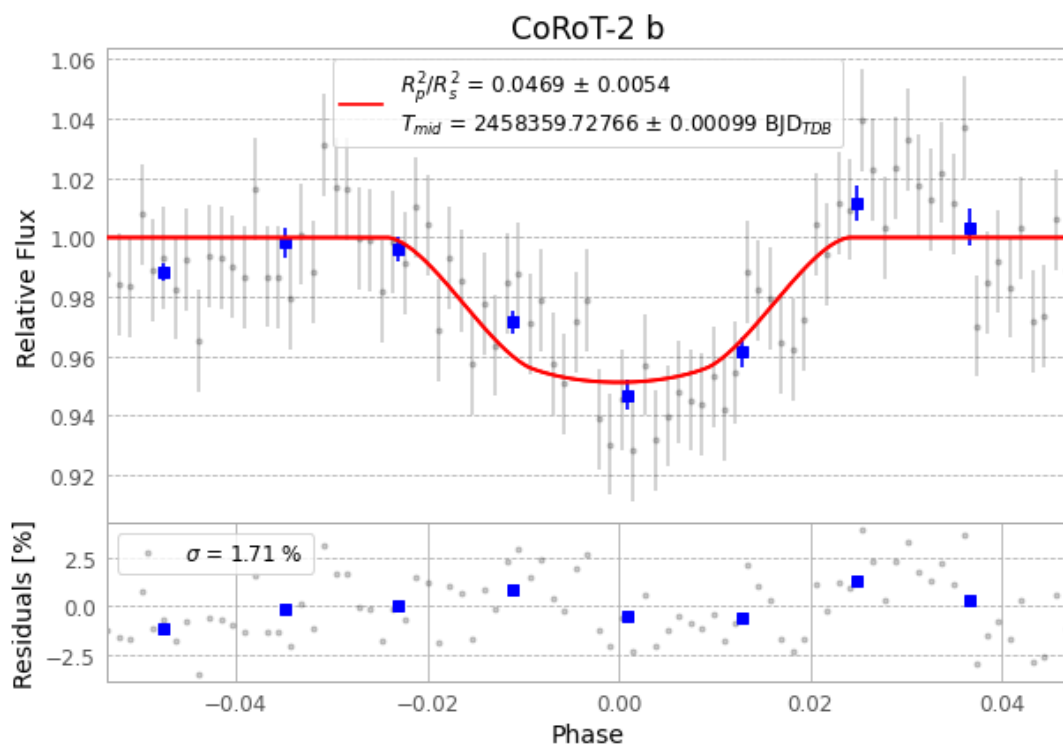


Figure 1 — FinalLightCurve_CoRoT-2 b_2018-08-28.png (SHA-256 6b8184539a6eba7b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [298, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf74b65ba7b8d9ac...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 378b18d

Data provenance

85 FITS frames (56 MB), CoRoT-2180829030612.FITS → CoRoT-2180829071812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-180829.log (1c36721a145d...), FinalLightCurve_CoRoT-2 b_2018-08-28.png (6b8184539a6e...), FinalLightCurve_CoRoT-2 b_2018-08-28.csv (f8148c322156...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 20:53Z.